

LABSHEET - 1**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24**

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
# 1. LOAD IMAGE
# =====

image_path = "/content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg"
image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Check the image path and filename.")
    exit()

print("Image loaded successfully!")
print("Himani Chaudhary")
print("CU240250967")
```

Image loaded successfully!
Himani Chaudhary
CU240250967

```
import cv2
import matplotlib.pyplot as plt

# Load image
image = cv2.imread("/content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg")

# Check image
if image is None:
    print("Error: Image not found. Check the file name/path.")
else:
    # Convert BGR to RGB
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # Display original image
    plt.figure(figsize=(6, 5))
    plt.imshow(image_rgb)
    plt.title("Original Image")
    plt.axis("off")
    plt.show()

print("Himani Chaudhary")
print("CU240250967")
```

Original Image



Himani Chaudhary
CU240250967

```
# =====
# 3. IMAGE PROPERTIES
# =====

if image is None:
    print("Error: Image was not loaded.")
else:
    height, width, channels = image.shape

    print("\n===== IMAGE PROPERTIES =====")
    print("Width      :", width)
    print("Height     :", height)
    print("Channels    :", channels)
    print("Resolution  :", f"{width} x {height}")
    print("Data Type   :", image.dtype)
    print("Total Pixels :", width * height)
    print("Total Values :", image.size)

print("Himani Chaudhary")
print("CU240250967")
```

```
===== IMAGE PROPERTIES =====
Width      : 5376
Height     : 3584
Channels    : 3
Resolution : 5376 x 3584
Data Type   : uint8
Total Pixels : 19267584
Total Values : 57802752
Himani Chaudhary
CU240250967
```

```
# =====
# 4. SAVE IMAGE IN JPEG AND PNG
# =====

if image is None:
    print("Error: Image is not loaded.")
else:
    cv2.imwrite("output_image.jpg", image)
    cv2.imwrite("output_image.png", image)

    print("\nImages saved successfully:")
    print("output_image.jpg")
    print("output_image.png")

print("Himani Chaudhary")
print("CU240250967")
```

```
Images saved successfully:
output_image.jpg
output_image.png
Himani Chaudhary
CU240250967
```

```
# 5. CONVERT TO GRAYSCALE
# =====

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

cv2.imwrite("grayscale.jpg", gray)
print("Himani Chaudhary")
print("CU240250967")
```

```
Himani Chaudhary
CU240250967
```

```
# 6. CONVERT TO HSV
# =====

hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

cv2.imwrite("hsv.jpg", hsv)
print("Himani Chaudhary")
```

```
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 7. CONVERT TO LAB
# =====

lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)

cv2.imwrite("lab.jpg", lab)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 8. RESIZE IMAGE
# =====

new_width = 400
new_height = 300

resized = cv2.resize(image, (new_width, new_height))

cv2.imwrite("resized.jpg", resized)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 9. ROTATE IMAGE
# =====

rotated = cv2.rotate(
    image,
    cv2.ROTATE_90_CLOCKWISE
)

cv2.imwrite("rotated.jpg", rotated)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 10. HORIZONTAL FLIP
# =====

horizontal_flip = cv2.flip(image, 1)

cv2.imwrite(
    "horizontal_flip.jpg",
    horizontal_flip
)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 11. VERTICAL FLIP
# =====

vertical_flip = cv2.flip(image, 0)

cv2.imwrite(
    "vertical_flip.jpg",
    vertical_flip
)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 12. IMAGE COMPLEMENT / NEGATIVE
# =====

negative = 255 - image

cv2.imwrite(
    "negative.jpg",
    negative
)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 13. CROP REGION OF INTEREST (ROI)
# =====

h, w = image.shape[:2]

# Define ROI
x1 = int(w * 0.25)
x2 = int(w * 0.75)

y1 = int(h * 0.25)
y2 = int(h * 0.75)

roi = image[y1:y2, x1:x2]

cv2.imwrite(
    "roi.jpg",
    roi
)
print("Himani Chaudhary")
print("CU240250967")
```

Himani Chaudhary
CU240250967

```
# 14. DISPLAY ALL PROCESSED IMAGES
# =====

plt.figure(figsize=(14, 10))

# Original
plt.subplot(3, 3, 1)
plt.imshow(
    cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
)
plt.title("Original")
plt.axis("off")

# Grayscale
plt.subplot(3, 3, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

# HSV
plt.subplot(3, 3, 3)
plt.imshow(
    cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB)
)
plt.title("HSV")
plt.axis("off")

# LAB
plt.subplot(3, 3, 4)
plt.imshow(
    cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)
)
```

```
plt.title("LAB")
plt.axis("off")

# Resized
plt.subplot(3, 3, 5)
plt.imshow(
    cv2.cvtColor(resized, cv2.COLOR_BGR2RGB)
)
plt.title("Resized")
plt.axis("off")

# Rotated
plt.subplot(3, 3, 6)
plt.imshow(
    cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB)
)
plt.title("Rotated")
plt.axis("off")

# Horizontal Flip
plt.subplot(3, 3, 7)
plt.imshow(
    cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB)
)
plt.title("Horizontal Flip")
plt.axis("off")

# Negative
plt.subplot(3, 3, 8)
plt.imshow(
    cv2.cvtColor(negative, cv2.COLOR_BGR2RGB)
)
plt.title("Negative")
plt.axis("off")

# ROI
plt.subplot(3, 3, 9)
plt.imshow(
    cv2.cvtColor(roi, cv2.COLOR_BGR2RGB)
)
plt.title("ROI")
plt.axis("off")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

Original



Grayscale



HSV



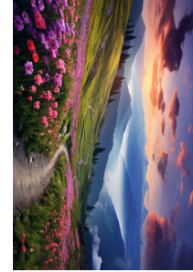
LAB



Resized



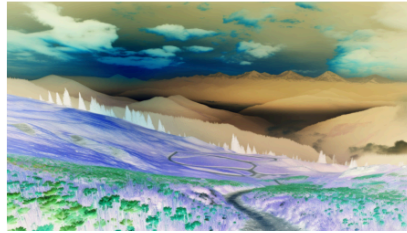
Rotated



Horizontal Flip



Negative



ROI

